

■ OpenClaw

Complete Setup & Reference Guide

Azure VM · Docker · WhatsApp Integration

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What is OpenClaw?

OpenClaw (openclaw.ai) is an open-source personal AI assistant that runs on your own machine or a cloud VM. Unlike hosted AI services, your context, memory and skills live on infrastructure you control. It connects to chat apps you already use — WhatsApp, Telegram, Discord, Slack — and can autonomously perform tasks: managing email, browsing the web, running shell commands, calling APIs, and building its own skills on the fly.

Why this setup?

Running OpenClaw in Docker on an Azure VM gives you the best of all worlds: OpenClaw gets full access to a dedicated machine it can 'own', your main Windows PC is completely untouched, and the VM can be deallocated when not in use to keep costs minimal.

1. Provision the Azure VM

Recommended Specification

Setting	Value
VM Size	Standard_B2s (2 vCPU, 4 GB RAM)
Estimated Cost	~£15/month (compute only)
OS Image	Ubuntu 24.04 LTS
Disk	30 GB Premium SSD
Region	UK South (or nearest)
Authentication	SSH public key

Azure CLI Commands

```
# Create resource group
az group create --name rg-openclaw --location uksouth

# Create the VM
az vm create \
  --resource-group rg-openclaw \
  --name vm-openclaw \
  --image Ubuntu2404 \
  --size Standard_B2s \
  --admin-username azureuser \
  --generate-ssh-keys \
  --public-ip-sku Standard
```

Note: When the VM is created with `--generate-ssh-keys`, Azure automatically adds a default-allow-ssh NSG rule at priority 1000. You do not need to add it manually — doing so will produce a `SecurityRuleConflict` error, which you can safely ignore.

Verify Current NSG Rules

```
az network nsg rule list \
  --resource-group rg-openclaw \
  --nsg-name vm-openclawNSG \
  --output table
```

SSH Into the VM

```
ssh azureuser@<your-vm-public-ip>
```

2. Install Docker

Install Docker on the VM using the official convenience script, then add your user to the docker group so you do not need sudo for every command.

```
# Install Docker
curl -fsSL https://get.docker.com | sudo bash

# Add user to docker group
sudo usermod -aG docker azureuser

# Apply without logging out
newgrp docker

# Verify
docker --version
docker compose version
```

3. Create OpenClaw Docker Files

Create a working directory on the VM and add the three files below. The easiest approach is VS Code Remote SSH — connect to the VM and create the files in the editor as you would locally. Alternatively use the cat heredoc commands shown below directly in your SSH session.

```
mkdir ~/openclaw && cd ~/openclaw
```

Dockerfile

```
FROM ubuntu:24.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && apt-get install -y \
    curl ca-certificates bash git \
    && curl -fsSL https://deb.nodesource.com/setup_22.x | bash - \
    && apt-get install -y nodejs \
    && rm -rf /var/lib/apt/lists/*

RUN npm install -g openclaw

ENV OPENCLAW_STATE_DIR=/data/state
ENV OPENCLAW_CONFIG_PATH=/data/config.json

WORKDIR /data
EXPOSE 18789

CMD ["openclaw", "gateway", "--port", "18789"]
```

docker-compose.yml

```
services:
  openclaw:
    build: .
    container_name: openclaw
    restart: unless-stopped

    ports:
      - "127.0.0.1:18789:18789" # Localhost only — access via SSH tunnel

    environment:
      - ANTHROPIC_API_KEY=${ANTHROPIC_API_KEY}

    volumes:
      - openclaw_data:/data # Persistent config & state
      - /:/host # Full VM filesystem access
      - /var/run/docker.sock:/var/run/docker.sock

    privileged: true # Full VM permissions
    network_mode: host # Same network as VM

volumes:
  openclaw_data:
```

.env

```
ANTHROPIC_API_KEY=your_api_key_here
```

Tip: Your Anthropic API key is at console.anthropic.com → API Keys. Add a payment method and top up with credits — personal OpenClaw usage typically costs \$0.50–\$2/day depending on activity.

Note: Note: privileged: true and the Docker socket mount give OpenClaw full control over the VM. This is intentional on a dedicated single-purpose VM. Do not install untrusted community skills without reviewing their source code first.

Verify Files

Hidden files (dotfiles like .env) are not shown by ls alone. Always use:

```
ls -la ~/openclaw
```

4. Build, Start & Configure OpenClaw

Build and Start the Container

```
cd ~/openclaw
docker compose up -d --build

# Follow the logs
docker compose logs -f
```

Troubleshooting: Container Keeps Restarting

If docker ps shows the container in a restart loop, check the logs:

```
docker compose logs --tail=50
```

A common cause is Missing config. Run openclaw setup — this means onboarding has not been completed yet. Fix it by running setup in a one-off container:

```
# Stop the restarting container
docker compose down

# Run setup interactively
docker compose run --rm openclaw openclaw setup

# Then bring it back up
docker compose up -d
```

Run the Onboarding Wizard

If the container is running, exec into it to run onboarding:

```
docker exec -it openclaw openclaw onboard
```

The wizard walks through: LLM provider & API key, model selection, and chat channel setup (WhatsApp, Telegram, Discord, etc.).

Find Your Gateway Token

If you need the gateway token at any point:

```
# If container is running
docker exec -it openclaw cat /data/config.json | python3 -m json.tool

# If container is stopped, read from the Docker volume directly
sudo cat /var/lib/docker/volumes/openclaw_openclaw_data/_data/config.json | python3 -m json.tool
```

Access the Control UI from Windows

Port 18789 is bound to localhost only on the VM. Use an SSH tunnel to access the Control UI securely without opening any firewall ports:

```
# Run this on your Windows machine (PowerShell or terminal)
ssh -L 18789:localhost:18789 azureuser@<your-vm-public-ip> -N

# Then open in your browser:
http://localhost:18789
```

Tip: If you use VS Code with the Remote SSH extension, it detects port 18789 and offers to forward it automatically — no manual tunnel command needed.

5. WhatsApp Integration

OpenClaw connects to WhatsApp by linking as a device to your existing personal WhatsApp account — the same way WhatsApp Web works. This is the cleanest approach as it requires no Twilio account or business API.

Setup During Onboarding

When you run `openclaw onboard` and select WhatsApp as a channel, OpenClaw displays a QR code. Scan it from WhatsApp on your phone:

- Open WhatsApp on your phone
- Tap the three-dot menu (Android) or Settings (iPhone)
- Select Linked Devices → Link a Device
- Scan the QR code shown in the terminal

Communicating with Your Agent

Once linked, send a message to yourself in WhatsApp (the Saved Messages / Message Yourself feature). OpenClaw listens on your account as a linked device and responds to messages you send to yourself.

Test Messages to Try

```
Hello, are you there?  
What can you help me with?  
Search the web for the latest .NET 9 release notes  
Remind me in 10 minutes to check my emails  
What is the weather in London today?
```

Cloudflare Tunnel

During onboarding OpenClaw may set up a Cloudflare tunnel to handle webhooks. This gives external services a public HTTPS URL to reach OpenClaw without you needing to open any ports on the Azure NSG — an elegant solution.

Note: WhatsApp linked device sessions can occasionally drop if your phone loses internet for a prolonged period. If OpenClaw stops responding on WhatsApp, a docker compose restart on the VM re-establishes the connection. Session data is persisted in the `openclaw_data` Docker volume.

6. Reference — Useful Commands

Docker — Day to Day

```
# Check what containers are running
docker ps

# Check all containers including stopped ones
docker ps -a

# View live logs
docker compose logs -f

# View last 50 log lines
docker compose logs --tail=50

# Stop OpenClaw (VM stays running)
docker compose stop

# Start OpenClaw again
docker compose start

# Restart OpenClaw
docker compose restart

# Shell into the running container
docker exec -it openclaw bash

# Full clean rebuild (wipes state – use with caution)
docker compose down -v && docker compose up -d --build
```

Azure VM — Cost Management

```
# Deallocate VM – stops compute billing (cheapest option when not in use)
az vm deallocate --resource-group rg-openclaw --name vm-openclaw

# Start the VM again
az vm start --resource-group rg-openclaw --name vm-openclaw

# Get current public IP (changes after each deallocate unless using static IP)
az vm show -d \
  --resource-group rg-openclaw \
  --name vm-openclaw \
  --query publicIps -o tsv
```

Note: The VM's public IP changes each time you deallocate and restart unless you assign a static public IP in Azure (~£3/month). If the changing IP becomes inconvenient, assign a static IP via the Azure Portal under the VM's Network Interface → IP Configuration.

Approximate Monthly Cost Breakdown

Item	When Running	When Deallocated
B2s Compute	~£15/month	£0

30 GB Premium SSD	~£3/month	~£3/month
Static Public IP	~£3/month	~£3/month (optional)
Anthropic API	~\$0.50–\$2/day	£0
Total (approx)	~£20–25/month	~£3–6/month

7. What to Explore Next

Suggested Things to Try on WhatsApp

- Ask it to remind you of something in X minutes — tests the scheduler
- Ask it to search the web for recent news on a topic — tests browser skills
- Ask it to write and run a bash script — tests full system access
- Ask it to summarise your emails — then connect Gmail for full integration

Gmail Integration

Connect your Gmail account so OpenClaw can monitor your inbox, summarise emails, and alert you via WhatsApp when a specific person emails you — your original use case. Set this up via the OpenClaw Control Panel under Channels / Integrations, or ask OpenClaw itself on WhatsApp: 'Can you connect to my Gmail?'

Skills & Community

- Visit clawhub.ai to browse community-built skills
- OpenClaw can write its own skills — just describe what you want it to do
- Join the Discord community at discord.com/invite/clawd for tips and help
- Monitor your Anthropic API spend at console.anthropic.com

Useful Links

Resource	URL
OpenClaw Website	openclaw.ai
Documentation	docs.openclaw.ai
GitHub	github.com/openclaw/openclaw
Community Skills Hub	clawhub.ai
Discord Community	discord.com/invite/clawd
Anthropic Console	console.anthropic.com
Azure Portal	portal.azure.com

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